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GABRIELE BELOTTI, PHD

Independent Consultant, X-ray and Medical Imaging Software & Systems

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WORK EXPERIENCE

Independent Consultant

Self-employed Milan (IT)

2026 - ongoing

- Consulting for companies and research centres developing **X-ray imaging systems**: **image quality** and **CBCT validation**, **tomographic reconstruction**, **geometric calibration** of robotic systems
- Instrumentation and embedded work; long-term **C++** codebase development and maintenance for imaging and industrial machinery

X-Ray Image-Processing/Software Engineer

Cefla s.c. Imola (IT)

Jan 2025 - 2026

R&D:

- **Image Quality** assurance via algorithms **parametrization**, phantom+clinical **data analysis**, **x-ray exposure** control, and **geometrical validation** techniques
- **After-sales** debugging and assistance for dental and clinical CBCT machines; **production** acceptance **testing** and anomaly **investigation**
- Flat Panel Detectors (FPDs) quality control and comparison using IEC standards
- Requirement and Functional **Design**, **back-end** implementation, cross-team **coordination**, **bug** fixing, and **continuous integration testing** in Gitlab for Neowise suite
- Legacy **tomographic code** maintenance, refactoring, and optimization

X-Ray Tomographic Reconstruction Software Engineer

medPhoton Salzburg (AT)

Sep 2023 - Dec 2024

Head of interdisciplinary Imaging Core Group (ICG):

- **Led a group** of 15 people towards **diagnosis and resolution** of **Image Quality** issues/customer complaints
- **Research** and **Design** of HW and SW **state-of-the-art** solutions
- Extraction and dissemination of **clinical data know-how**
- Consultation and assistance to Medical Physicists, Clinicians, Software Engineers, Test Engineers, and Product managers

R&D:

- Design and perform **Experimental activities** for prototypes: optical tracking, flexmap calibration, automatic exposure control, room robotic navigation and stereotaxis, MLG detector calibration, Factory Acceptance Test verifications

TECHNICAL SKILLS

Computer literate · LaTeX · Open Source · Linux · Agile · MS Windows · System integration

C++ · C · CUDA · Python · Git · Medical Images · OpenCV · DICOM · ITK · VTK · RTK · CMAKE · OpenGATE · Geant4 · Slurm · Jira · Embedded Systems · MATLAB/Simulink · Computer-Assisted Surgery · Image Registration · Segmentation

LANGUAGES

Italian ●●●●●

English ●●●●●

French ●●○○○

EDUCATION

Ph.D. in Bioengineering

Politecnico di Milano

Nov 2019 - Mar 2024 cum Laude

Development of technologies for in-room image guidance in hadron therapy

M.Sc. in Biomedical Engineering

Politecnico di Milano

Oct 2016 - Apr 2019 106 /110

Image-guided estimation of the grasping tool pose for robotic-assisted surgical needle insertion

B.Sc. in Biomedical Engineering

Politecnico di Milano

Oct 2013 - Sep 2016 93 /110

Table bound play robot for motion impaired children

- Led **tomographic code** implementation, maintenance, refactoring, and optimization (∼5 people)
- **Post Processing filters** implementation and testing on test machines
- Responsible for **software documentation** for the Cone Beam CT Module (CBCT-M)

PhD Researcher in Computer-Aided Radiation Therapy

Politecnico di Milano Milan (IT)

May 2019 - Aug 2023

Design and development of a novel custom robotic C-arm for X-Ray In-Room imaging for CNAO particle therapy center:

- **Full-stack C++ development** for X-ray Patient Positioning Verification suite
- Design, implementation, and verification of a 9 degrees of freedom **Geometrical calibration (Flexmap approach)**
- **Integration** with the treatment room infrastructure and **Clinical Commissioning**
- **Tomographic reconstruction** in **Cone Beam** geometry
- **Academic writing**; **oral presentation** at CT-meeting 2022

Teaching Assistant at Politecnico di Milano (3 M.Sc. courses)
Electronics Technologies and Biosensors Laboratory, *Computer-Aided Surgery*, *Laboratory of Radiotherapy Technologies*
Thesis supervision: four M.Sc. and B.Sc. theses on adaptive radiotherapy, 4D CBCT, and CBCT scatter reconstruction and correction

Visiting PhD Researcher

INSA Lyon - CREATIS lab Lyon (FR)

Nov 2021 - Apr 2022

Monte Carlo simulations for particle transport in Cone Beam Computed Tomography

- **Open source** contributions (RTK, OpenGATE)
- **Distributed computing** (**Linux** based **Slurm** platform for **in2p3**)
- **Scatter simulation** and correction
- Development of novel acquisition geometries

Internship - Research Engineer

ICube Lab Strasbourg (FR)

Sep 2017 - Mar 2018

Topics: Coding (C++, MATLAB); X-Ray imaging; Computer Vision; Robot Assisted Interventional Radiology

RESEARCH TOPICS

Imaging Ring Mobile (IRm)

Sep 2023 - Dec 2024

Scatter correction; dual arc CBCT and saddle trajectories; dual energy CBCT; 3D subtraction angiography

Robotic C-arm CBCT system for CNAO lateral room

Nov 2019 - June 2023

Cone-beam CT reconstruction and FOV extension; adaptive radiotherapy strategies in hadron therapy; time-correlated (4D) in-room CBCT; X-ray and optical in-room imaging for patient positioning verification; deep learning for artifact reduction and image quality improvement in CBCT

PUBLICATIONS

Journal Articles

- Belotti, Gabriele, Giovanni Fattori, et al. (2024). “Extension of the cone-beam CT field-of-view using two complementary short scans”. In: *Medical Physics* 51.5, pp. 3391-3404.
- Knäusl, Barbara et al. (2024). “A review of the clinical introduction of 4D particle therapy research concepts”. In: *Physics and imaging in radiation oncology* 29, p. 100535.
- Rossi, Matteo, Gabriele Belotti, Luca Mainardi, et al. (2024). “Feasibility of proton dosimetry overriding planning CT with daily CBCT elaborated through generative artificial intelligence tools”. In: *Computer Assisted Surgery* 29.1, p. 2327981.
- Belotti, Gabriele, Matteo Rossi, et al. (2023). “A new system for in-room image guidance in particle therapy at CNAO”. In: *Physica Medica* 114, p. 103162.
- Antonioli, L. et al. (2021). “Convolutional Neural Networks Cascade for Automatic Pupil and Iris Detection in Ocular Proton Therapy”. In: *Sensors* 21.13, p. 4400.
- Ricotti, Rosalinda et al. (2021). “Dosimetric effect of variable rectum and sigmoid colon filling during carbon ion radiotherapy to sacral chordoma”. In: *Physica Medica* 90, pp. 123-133.
- Rossi, Matteo, Gabriele Belotti, Chiara Paganelli, et al. (2021). “Image-based shading correction for narrow-FOV truncated pelvic CBCT with deep convolutional neural networks and transfer learning”. In: *Medical Physics* 48.11, pp. 7112-7126.

Conference Proceedings

- Russo, F. M. et al. (2024). “Enhancing CBCT image quality using a novel sCT-based shading correction”. In: *XXth International Conference on the Use of Computers in Radiation Therapy (ICCR 2024)*. Lyon, France.
- Belotti, Gabriele, Simon Rit, and Guido Baroni (2022). “Extension of the cone-beam CT field-of-view using two short scans with displaced centers of rotation”. In: *Seventh International Conference on Image Formation in X-Ray Computed Tomography XCT22*. Vol. 12304. SPIE.